Design Quiz
08 Substitution Cipher

Technical Requirements

List the technical requirements you have identified in the instructions. Remember, these are requirements that need to be met to earn full autograder credit, but that don't affect the program output.

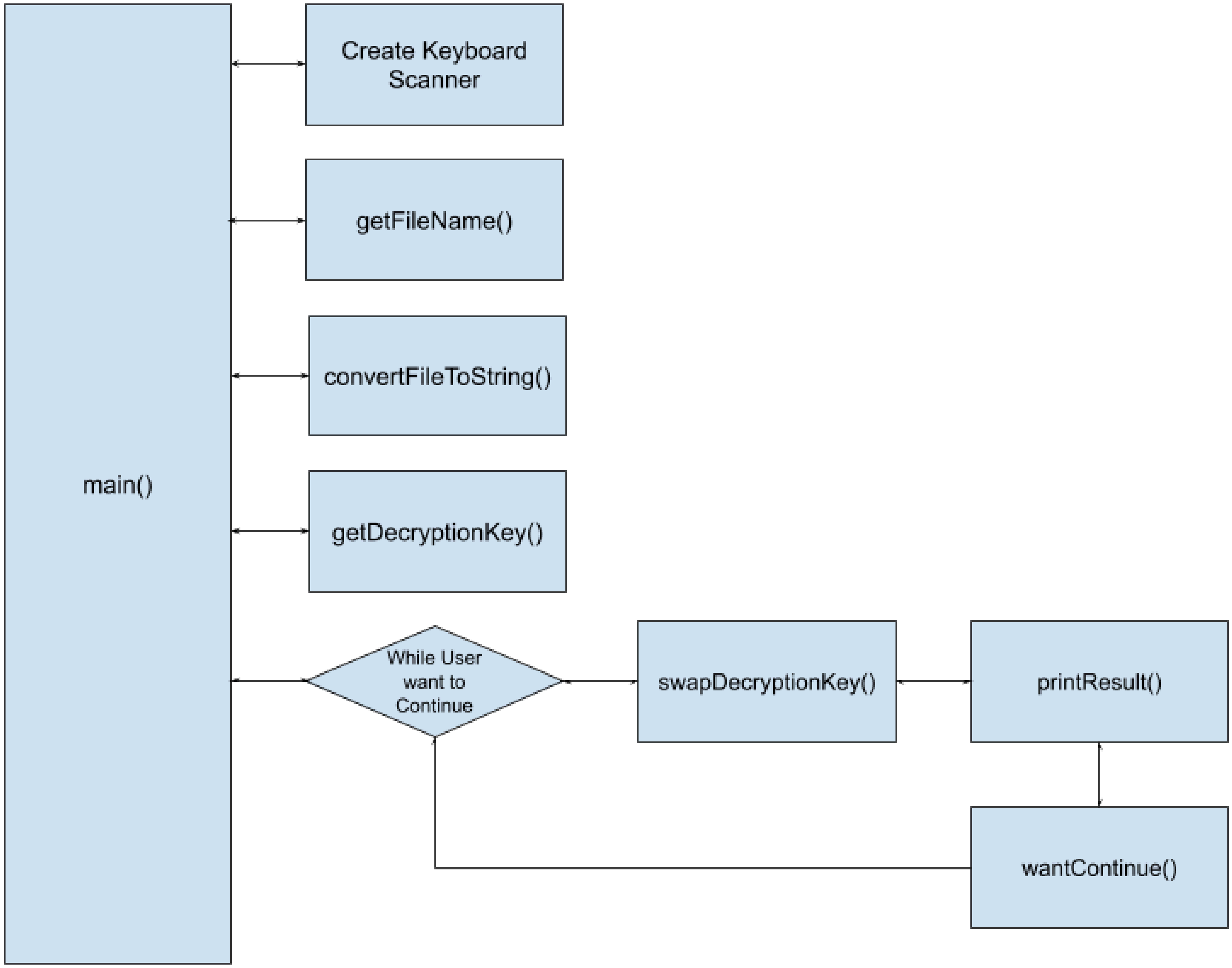
- Only use a 1D Array
- You can't use class variables
- No 2D or greater arrays
- You can't convert the decryption key char array into a String.
- Do not change the starter code
- Use the ASCII characters only within the range of [32, 126].
    - Even though the table is 128 elements long
- Asks the user whether want to continue swapping one character for another character
    - If they say yes, then you swap the character in the key array at the integer index of the original character.
- Use the given keyboard scanner only, do not create another Scanner

High-Level Summary

Write a high-level summary of the solution in English words. This should correspond, more or less, to the main() method.

- public static void main(String[] args) {
    - Create Keyboard Scanner
    - Call getFileName()
    - Call convertFileToString() method from the DecryptUtilities file
    - Use the get decryption key method
    - Create the Frequency char array
    - Loop while the user wants to continue to swap
        - Call the Swap decryption key method
        - Call the print out result method.
        - Ask to see if they want to continue

Program Flow:



Variables and Data Structures

Specify how key data will be stored in your program.

Variable Type and/or Data Structure IF AN ARRAY - Indicate the meaning of the index and the value.	Data
Public static final int SPACE = 32; Public static final int TILDE = 126; Public static final int ASCII_LEN = 128;	space, tilde and number of ASCII characters, to avoid magic numbers
int[] charFrequencies = new int[ASCII_LEN] - Index: Each index represents the int value of the char at that index - Value: The value represents the	character frequencies

number of time the char appears int the file.	
char[] keySwitchList = new char[ASCII_LEN] - Index: Each index represents the original list of characters from the - Value: the Char that the original	decryption key representing the best guess for which characters should replace which other characters